Experiment 5: Containerizing ML API using Docker with Cloud Registry and Kubernetes

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AIM

To study containerization of an ML API using Docker and its deployment to cloud registries and Kubernetes clusters.

OBJECTIVES

1. To understand containerization using Docker.
2. To study offline setup with Docker Desktop for local testing.
3. To understand deployment to cloud registries and Kubernetes for scalability.

THEORY

1. Containerization using Docker

Containerization packages an application with all its dependencies into a portable container.

Docker is used to build and manage these containers. In ML APIs, Docker ensures consistent runtime environments across systems, improving portability and reliability.

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2. Offline Setup: Docker Desktop

Docker Desktop is used to build and test container images locally. The ML API is verified in a local container before cloud deployment.

This step ensures proper functionality and reduces deployment errors.

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3. Cloud Container Registries

After testing, images are pushed to cloud registries such as:

- Azure Container Registry
- GCP Artifact Registry

These services store container images securely and enable integration with cloud deployment platforms.

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4. Deployment using Kubernetes

For scalable hosting, containers are deployed to Kubernetes clusters like:

- Azure Kubernetes Service
- Google Kubernetes Engine

Kubernetes manages scaling, load balancing, and reliability, making ML APIs production-ready.

CONCLUSION

In this experiment, we studied Docker-based containerization of an ML API, local testing, and cloud deployment using registries and Kubernetes.

This workflow ensures portability, scalability, and efficient ML application deployment.